Shreyas Girish

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Education

- **Universität Politècnica de Catalunya** — Barcelona, Spain
  Master of Numerical Methods in Engineering — *Sept 2022 - May 2025*
  Courses: *Advanced Fluid Mechanics, HPC, Machine Learning, Artificial Intelligence, Structural Dynamics, Parametric Model Order Reduction(PMOR)*
- **JSS Academy of Technical Education** — Bangalore, India
  Bachelor of Engineering - Mechanical Engineering — *Aug 2015 - June 2019*
  Courses: *Fluid Dynamics, FEA, Thermodynamics, Heat Transfer, Turbo Machinery*

Skills Summary

- **Modeling**: CATIA, Solidworks, Spaceclaim, HyperMesh, AutoCAD
- **CAE Tools**: ANSYS, Star-CCM+, Abaqus, OpenFoam, Fluent, Workbench
- **Programming Languages**: Python, C++, MATLAB, TCL/TK
- **Languages**: **Native/ Bilingual Proficiency** - English, **Limited Proficiency** - Spanish, German
- **Soft Skills**: Leadership, Task Management, Writing, Public Speaking, Time Management

Work Experience

- **Mekano4** — March 2025 - May 2025
  Intern
  - **Bridge Component Simulation:**: Executed nonlinear structural simulations using ANSYS to assess the behavior of post-tensioned systems, incorporating geometric and material nonlinearity for critical bridge elements.
  - **Contact & Convergence Modeling:**: Developed and refined models with complex contact interactions (wedge–anchor–cable interfaces), implementing advanced solver strategies to ensure accurate convergence and displacement control.
- **CSolar.es** — Oct 2024 - Dec 2024
  Intern
  - **Solar Panel CFD Analysis**: Conducted comprehensive CFD simulations to analyze airflow performance over solar panels, optimizing efficiency and durability.
  - **Structural Simulation Expertise**: Performed advanced structural analyses using Abaqus to evaluate and enhance the structural integrity of solar panel systems.
  - **Industry Presentation**: Presented key findings and technical insights at an ANSYS event, demonstrating expertise in simulation methodologies and their real-world applications.
- **Faurecia Clean Mobility** — June 2022 - Sep 2022
  Senior CFD Engineer - Full Time
  - **Exhaust System Optimization**: Led team to improve exhaust system simulation efficiency using CAE tools and advanced programming.
  - **Parametric & Comparative Analysis**: Performed parametric studies and comparative analysis to enhance simulation accuracy and insights.
  - **Multidisciplinary Collaboration**: Scoped projects with field engineers, delivering professional services while ensuring high-quality client support.
- **Copes Tech India Pvt. Ltd** — Jan 2020 - May 2022
  CFD Analyst - Full Time
  - **CFD & Thermal Analysis**: Conducted fluid flow and thermal CFD analysis of engine exhaust systems, with Star-CCM+ and Hypermesh.
  - **Simulation Optimization**: Led geometry optimization, mesh modeling, and structural dynamics simulations.
  - **Advanced Simulation Tools**: Utilized Star-CCM+, Hypermesh, and Ansys to develop comprehensive solutions, optimize designs, and enhance functionality and efficiency.
- **Copes Tech India Pvt. Ltd** — Jul 2019 - Jan 2020
  CFD Engineer - Full Time
  - **Advanced CFD Simulation**: Executed 30+ 1D, 2D, and 3D simulations, showcasing expertise in fluid flow and thermal analysis with high accuracy and efficiency.
  - **Technical Proficiency**: Demonstrated advanced skills in modeling, meshing, and simulation optimization using tools like Star-CCM+, Hypermesh, and Ansys.

- **BHEL-EDN** Nov 2018 - Dec 2018
  Intern
  - **Cold Plate Design & Testing**: Designed and rigorously tested a cold plate for traction applications, showcasing innovation and practical problem-solving capabilities.
  - **Advanced Modeling**: Created a detailed SolidWorks model to support comprehensive analysis and enhance traction system performance.
  - **Thermal Management Expertise**: Applied engineering principles to optimize cold plate design, ensuring efficient heat dissipation and reliability in traction systems.
- **Triveni Engineering & Industries Limited** Jul 2018 - Aug 2018
  Intern
  - **Gear Manufacturing Expertise**: Gained hands-on experience in the complete gear manufacturing process, including raw material selection, production, and testing, with a focus on practical engineering applications.

Projects

- **3D Modeling of Automotive Suspension and Seating Systems (CATIA V5)** *Tools: CATIA V5 (Part Design, Assembly Design, Drafting)*
  - **Designed Suspension Components -** Modeled wishbones, uprights, coil springs, and damper assemblies with accurate kinematics.
  - **Modeled Seating Systems -** Created seat frames, brackets, and ergonomic profiles using Part and Surface workbenches.
  - **Assembly & Motion Checks -** Assembled components ensuring clearance and fitment validation.
  - **Created Manufacturing Drawings -** Generated 2D GD&T annotated drawings with exploded views.
  - **Performed Interference Analysis -** GConducted clash detection for production feasibility.
- **Gearbox Housing Design and Drafting for Industrial Applications (SolidWorks)** *Tools: SolidWorks (Part Modeling, Assembly, Drafting, SimulationXpress)*
  - **Modeled Gearbox Housing** – Designed internal mounting features for shafts, bearings, and gears.
  - **Drafting Documentation** – Generated exploded views, BoM, and manufacturing drawings.
  - **Developed Variant Configurations** – Automated gearbox variants using Design Tables.
  - **Automated Repetitive Modeling** – Created SolidWorks macros for task automation.
- **Structural Simulation of Bridge Bearings and Anchor Systems (ANSYS Workbench)** *Tools: ANSYS Workbench (Static Structural, Contact Analysis)*
  - **Simulated Bearings and Anchors -** Evaluated stress distribution under various loading conditions.
  - **Modeled Contact Interactions -** Simulated frictional and contact behavior for realistic load transfer.
  - **Performed Mesh Convergence Studies -** Ensured simulation accuracy while optimizing resources.
  - **Design Optimization Insights -** Provided structural improvement recommendations.
  - **Project executed at -** *Mekano4 Ingeniería*, Barcelona.
- **Buckling Analysis of Solar Panel Support Structures (Abaqus CAE)** *Tools: Abaqus (Static Analysis, Linear Buckling)*
  - **Conducted Buckling Simulations -** Analyzed support structures under wind-induced lateral loads.
  - **Defined Realistic Load Scenarios -** Modeled boundary conditions reflecting field installations.
  - **Captured Critical Stress Zones -** Applied mesh refinements near joint regions.
  - **Validated with Euler's Theory -** Provided buckling optimization strategies.
  - **Project performed for -** *CSolar*, Barcelona.
- **CFD Analysis of HVAC Airflow Distribution in Vehicle Cabin (OpenFOAM)** *Tools: OpenFOAM, Python, RANS Models (k-epsilon, k-omega)*
  - **Simulated Cabin Airflow -** Optimized HVAC vent placement for enhanced passenger comfort.
  - **CAD to CFD Workflow -** Developed 3D cabin models using CATIA V5 and imported into OpenFOAM.
  - **Automated Mesh Refinement -** Captured flow recirculation and dead zones.
  - **Analyzed Velocity and Thermal Field -** Evaluated airflow exposure and cooling efficiency.
  - **Validated with Experimental Data -** Ensured model accuracy through test comparisons.

- **Supersonic Flow Simulation in Converging-Diverging Nozzles (MATLAB)** *Tools: MATLAB, MacCormack Method, Compressible Flow Analysis*
  - **Developed Solver for Compressible Flow -** Simulated supersonic nozzle flow using 1D MacCormack method.
  - **Analyzed Shock Wave Formation -** Studied flow variations across nozzle geometries.
  - **Validated with Theoretical Models -** Compared outputs with isentropic and shock relations.
  - **Provided Thrust Optimization Insights -** Evaluated effects of geometry on thrust performance.
- **Master Thesis: Data-Driven Reduced Order Modeling for Nonlinear Structural Dynamics** *Tools: Python, NumPy, SciPy, KratosMultiphysics, Operator Inference, SVD*
  - **Developed Non-Intrusive ROM -** Implemented Operator Inference for nonlinear structural dynamics.
  - **Full-Order Simulation Dataset Generation -** Conducted high-fidelity simulations in KratosMultiphysics.
  - **Applied SVD for Dimensionality Reduction -** Achieved over 98% accuracy in reduced models.
  - **Validated ROM Performance -** Compared predictions against full-order model responses.
  - **Proposed Scalable ROM Framework -** Designed methodology for large-scale industrial applications.

Publications

- **A Literature Study on Automobile Fuel/Propulsion System Technologies**: Published on: 31 July 2017. ISSN: 2320-5547
- **A Literature Survey on Automobile fuel system technologies**: Published on: 31 July 2017. ISSN: 2320-5547
- **National Conference on Emerging Trends in Material, Design, and Manufacturing (NCMDM- 2019)**: Characterization of E Glass Fiber reinforced with Aluminum 8176 metal matrix composite, B.E. Project

Driving License

- **Driving License:** Type B

Relocation

- **Open to relocation across Europe and internationally**